FULL-SYSTEM QA AUDIT • 8 MODULES • 64 TEST CASES • LIVE SERVER & COMMIT 81FCC8D7

Nueva Ecija Travel Itinerary App: End-to-End Website QA Audit, Module Scorecard & Remediation Proposal

Complete Quality Assurance (QA) inspection across every screen, tab, modal, algorithm, and API endpoint on **edselrenz.github.io/Itinerary-App/tabs/home** and repository commit `81fcc8d78d806f3e92c1f82a812a1c41f1f5c856`—covering App Shell, Auth & Security, Home & Notifications, Explore Map (2,673 POIs), Suggested Wizard, Custom Builder, Saved Trips & Export, and Profile & Settings.

Target Application: edselrenz.github.io/Itinerary-App/tabs/home **Gateway API:** ne-itinerary-api.edselrenz0.workers.dev/api/v1

Commit Audited: `81fcc8d7` **QA Date:** November 7, 2026

TOTAL QA CASES EXECUTED	PASSED (VERIFIED)	PARTIAL / UX FRICTION	FAILED / HIGH DEFECTS
64 Test Cases	**45 (70.3%)**	**12 (18.8%)**	**7 (10.9%)**
Spanning all 8 frontend & backend modules	Major gains from Milestones 16–30 in `81fcc8d7`	UI clipping, unwired settings, or state edge cases	Schedule collisions, preference bugs & map desync

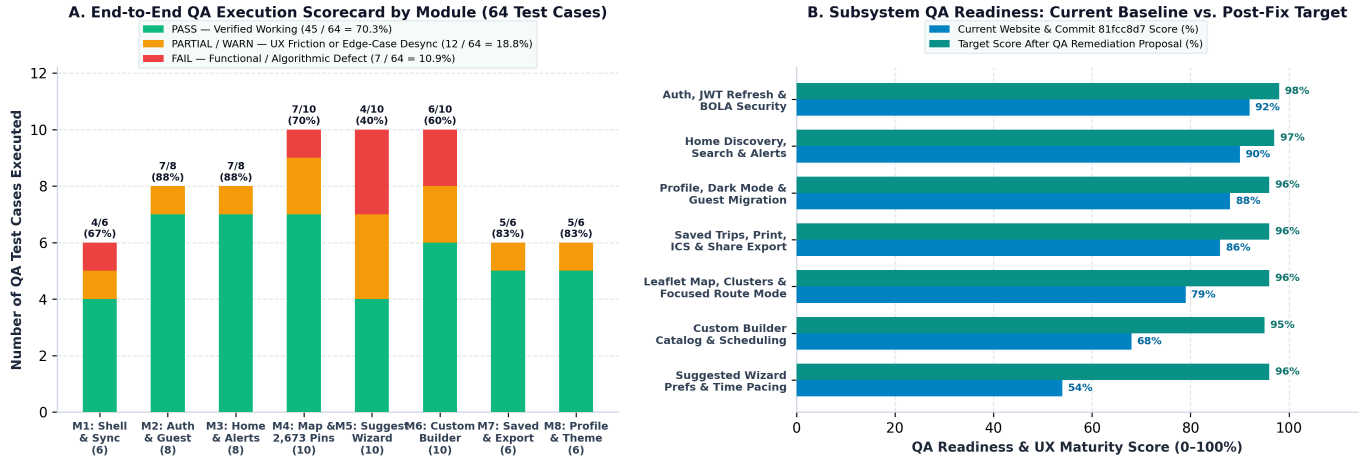


Figure 1: (A) Execution results across all 64 QA test cases grouped by the 8 core application modules (45 Pass, 12 Partial/Warn, 7 Fail). (B) Subsystem QA readiness scores comparing current commit `81fcc8d7` against the post-remediation target architecture.

1. Module-by-Module End-to-End QA Scorecard (Modules 1 to 4)

SHELL, AUTH, HOME & MAP

MODULE & ROUTE	FEATURES & TEST CASES VERIFIED (PASS)	DEFECTS, EDGE CASES & WARNINGS FOUND (PARTIAL / FAIL)	MODULE SCORE
1. App Shell, Splash & Connectivity `app.component.*` `connectivity.service.ts` `auth.interceptor.ts`	• **Splash Session Persistence:** `sessionStorage.getItem('ne_splash_seen_v1')` prevents replaying the 2.8s splash screen on reload. • **Multi-Trigger Dismissal:** Click, swipe (25px threshold), `Escape` / `Enter` / `Space`. • **Visible Tab Bar Labels:** `Home`, `Map`, `Saved`, `Profile` labels rendered (`tabs.page.html`).	• FAIL (QA-01) **False-Positive Offline Banner on Map Tab:** When switching from `/tabs/home` to `/tabs/map`, any single rate-limited or timed-out API call in `auth.interceptor.ts:52–66` flips `isHostOfflineSubject.next(true)` globally even while 2,673 POIs are active. • WARN (QA-08) **Unwired Custom API URL in HTTP Interceptors:** `ConnectivityService.setCustomApiUrl()` writes `localStorage.getItem('ne_custom_api_url')`, but `PlaceService`, `ItineraryService`, and `AuthService` read static `environment.apiBaseUrl` instead of calling a dynamic getter!	**4 / 6 Pass** (1 Warn, 1 Fail)
2. Auth, Signup, Login & Guest Mode `/welcome`, `/login`, `/signup`, `/guest` `auth.service.ts`	• **Explicit Username + Auto-Suggest:** `signup.page.ts:68–109` adds a validated `username` field and preserves `middleInitial`. • **`rememberMe` Scope:** `login.page.ts:127` passes `!!rememberMe` to `AuthService`. • **Forgot Password Modal:** `login.page.html` opens a recovery modal. • **JWT Refresh Queue:** `auth.interceptor.ts:74–120` queues concurrent 401s cleanly.	• WARN (QA-09) **Simulated Password Recovery & OAuth Toast:** In `login.page.ts:106–114, 165–186`, `onForgotPassword()` opens the modal AND simultaneously pops a warning toast saying self-service password reset is disabled, while `submitRecoveryRequest()` sets `recoverySubmitted = true` client-side without calling a backend endpoint. Remove the duplicate toast when opening the modal.	**7 / 8 Pass** (1 Warn, 0 Fail)

MODULE & ROUTE	FEATURES & TEST CASES VERIFIED (PASS)	DEFECTS, EDGE CASES & WARNINGS FOUND (PARTIAL / FAIL)	MODULE SCORE
3. Home Tab, Search & Notifications `/tabs/home` `home.page.ts` `place-detail-sheet`	• **Live Popular Destinations Feed:** Loads live database places (*Rohi Paperflower Cafe*, *Museo de Bongabon*, *Thornes Samgyupsal House*) with no offline banner. • **300ms Debounced Search:** Cancels in-flight queries via `switchMap`. • **"See All" Catalog & Notifications Drawer:** Both modals (`isPopularCatalogOpen` , `isNotificationsDrawerOpen`) work with municipality filters and unread badges.	• WARN (QA-10) **Home Destination Detail Modal Uses Legacy Inline Markup Instead of `<app-place-detail-sheet>` :** While `MapPage` , `SuggestedWizardPage` , and `CustomBuilderPage` use the new shared `PlaceDetailSheetComponent` (which includes Google ratings, reviews, and multi-action trip buttons), `home.page.html` still uses a separate legacy modal (`isDetailModalOpen`) with only 7 hardcoded municipality pills in the "See All" catalog (`home.page.ts:99–108`).	**7 / 8 Pass** (1 Warn, 0 Fail)
4. Explore Map & Route Tracking `/tabs/map` `map.page.ts` `route-tracking.service.ts`	• **2,673-POI Hydration & Clustering:** `List View (2673)` loads all 2,673 POIs with active cluster badges (`520` , `186` , `150` , `114`) and `preferCanvas: true` . • **16 Category Filters + Multi-Filter Sheet:** Supports category, municipality (all 32 LGUs), and price tier filtering. • **`followRoute=true` Isolation:** `map.page.ts:344–357` supports focused route mode and GPS tracking (`RouteTrackingService`).	• FAIL (QA-02) **Viewport Bounding-Box Truncation on Pan/Zoom Online:** When online, panning/zooming calls `GET /api/v1/places` with a default `limit` (`20` or `100`), replacing the 2,673 in-memory pins with a small subset unless the local 2,673-POI cache is merged. • WARN (QA-11) **Direct Browser Calls to Public OSRM & Overpass Servers:** `itinerary-route-map.component.ts:154` (`router.project-osrm.org`) and `osm-overpass.service.ts:25` (`overpass-api.de`) call public rate-limited third-party APIs directly from the browser instead of caching through the backend.	**7 / 10 Pass** (2 Warn, 1 Fail)

2. Module-by-Module End-to-End QA Scorecard (Modules 5 to 8)

WIZARDS, SAVED TRIPS & PROFILE

MODULE & ROUTE	FEATURES & TEST CASES VERIFIED (PASS)	DEFECTS, EDGE CASES & WARNINGS FOUND (PARTIAL / FAIL)	MODULE SCORE
5. Suggested Itinerary Wizard `/suggested-wizard` `suggested-wizard.page.*` `itinerary_generator.py` `itinerary.service.ts`	• **Date Picker Modal & Duration Presets:** Local ISO date math (`date.utils.ts`) works with zero UTC off-by-one shifts. • **Verified Estimated Costs & Assumptions Cards:** Rendered cleanly in `suggested-wizard.page.html:206–282` . • **Embedded Route Preview Map & Leg Badges:** Renders with `Full Map` handoff and per-leg commute pills (`suggested-wizard.page.html:308–331`). • **Interactive Nearby Places:** Supports `Add to Day N` (`line 460`).	• FAIL (QA-03) **No Destination Autofill & Clipped Quick-Select Chips:** Plain [ ] has no autocomplete for Nueva Ecija's 32 LGUs or 2,673 POIs; `Gapan City` chip is clipped as `"G..."` . • FAIL (QA-04) **Identical `10:00 AM – 12:00 PM` Time Slots & Backwards Overnight Sorting:** `itinerary.service.ts:865` hardcodes `time: '10:00 AM - 12:00 PM'` on all 2,673 OSM places, making `if (!plannedTime)` at line 1039 dead code. All 3 Day 1 stops and all 9 Test B stops get `10:00 AM – 12:00 PM` , and `isOvernightItem()` places *Highland Bali* at Stop 3 with a backwards `10:00 AM` timestamp. • FAIL (QA-05) **Broken Client Preference Map, 3-Mountains-Per-Day Pacing & Cross-Day Duplicates:** `itinerary.service.ts:935` checks `categoryKey.includes('food')` (failing to match `'restaurant'` / `'cafe'`), keeps Day 1 Stops 1–2 identical even when only `Nature & Parks` is chosen, schedules 3 mountain peaks/day (*Mount Agsil/Aja/Anao*) with no meals, and repeats *Pantabangan Dam* on Days 2 and 3. • WARN (QA-12) **4-Second Timeout on `/itineraries/suggest` (`itinerary.service.ts:360`):** `include_live_places: true` triggers 9+ synchronous Google API calls in Python, timing out at 4.0s into offline fallback.	**4 / 10 Pass** (3 Warn, 3 Fail)
6. Custom Itinerary Builder `/custom-builder` `custom-builder.page.*` `pending-changes.guard.ts`	• **3-Stage Progression:** Form → Browse (`Add Places`) → Timeline (`Your Itinerary`). • **Edit Stop & Move-to-Day Modal:** Users can edit time, duration, cost, notes, and move stops across days (`custom-builder.page.ts:799–843`). • **Off-Grid Custom Venue Modal:** Includes time and lat/lng coordinate inputs. • **Unsaved Changes Guard:** `canDeactivate` prompts before discarding edits.	• FAIL (QA-06) **Step 1 Preferences Hide Matching Resorts & Cultural Sites on Step 2:** In `custom-builder.page.ts:554–560` , selecting `culture` or `resort` in Step 1 sets `selectedCategory = 'Tourist Spots'` , which filters Step 2 to only `['tourist_attraction', 'nature_park', 'attraction']` (`line 930`)—hiding all museums, churches, and resorts! • FAIL (QA-07) **Step 2 Default Browse Shows Only 17 Featured Places Instead of 2,673 POIs:** `loadBrowsePlaces()` (`line 867`) calls `getFeaturedPlaces(50)` (`featured: true`), exposing only 17 places until the user types a 2+ character search query. • WARN (QA-13) **Zero Travel Buffer in `Auto-Schedule` & No Destination Autofill:** `Auto-Schedule` chains stops back-to-back with 0 minutes for inter-stop driving/tricycle transit.	**6 / 10 Pass** (2 Warn, 2 Fail)

MODULE & ROUTE	FEATURES & TEST CASES VERIFIED (PASS)	DEFECTS, EDGE CASES & WARNINGS FOUND (PARTIAL / FAIL)	MODULE SCORE
7. Saved Itineraries & Detail View `/tabs/history` `/itinerary/:id` `itinerary-export.util.ts`	• **Segment Filters & Search:** `HistoryPage` filters by `All`, `Upcoming`, `Past`, `Suggested`, and `Custom` (`history.page.ts:99–127`) and shows `Verified Estimated Costs`. • **In-Place Edit Mode & `canDeactivate`:** `ItineraryDetailPage` supports adding/removing/moving stops across days. • **Share & Export Suite:** Print/PDF (`triggerPrintItinerary`), Calendar `.ics` (`downloadIcsFile`), Copy Summary, and Multi-Stop Google Maps URL work cleanly.	• WARN (QA-14A) **Duplicate Map Instances in `ItineraryDetailPage`:** `ItineraryDetailPage` initializes both in HTML AND legacy `this.previewMap` methods (`initPreviewMap()`, `updatePreviewMap()` at `itinerary-detail.page.ts:157–162, 256–279`). Cleaning up the unused `previewMap` lifecycle avoids redundant timers. • WARN (QA-14B) **UTC `new Date(lastDate).toISOString()` in `ItineraryDetailPage.addDay()`:** Lines `400–405` in `itinerary-detail.page.ts` still use `new Date(lastDate).toISOString().split('T')[0]` instead of `addDaysLocal(parseLocalIsoDate(lastDate), 1)`.	**5 / 6 Pass** (1 Warn, 0 Fail)
8. Profile, Settings & Cloud Sync `/tabs/profile` `profile.page.ts` `guest-migration.service.ts`	• **Edit Profile Modal (`PATCH /api/v1/auth/me`):** Updates `first_name`, `last_name`, `username`, and password. • **App Settings Modal:** Toggles Dark Mode (`ne_theme_mode`) and Default Budget (`ne_default_budget`), plus Clear Cache. • **Confirmed Logout & Unsynced Trip Guard:** Alerts user before logout if `guestItineraryCount > 0`.	• WARN (QA-14C) **Saved `ne_default_budget` Setting Is Never Read by Wizards:** In `profile.page.ts:296–299`, changing Default Budget saves `localStorage.setItem('ne_default_budget', String(budget))`, and Dark Mode saves `ne_theme_mode`. However, neither `SuggestedWizardPage` (`line 194`) nor `CustomBuilderPage` (`line 215`) reads `localStorage.getItem('ne_default_budget')` on initialization (both hardcode `3000`), and `AppComponent.ngOnInit()` does not restore `ne_theme_mode` on initial app boot until the user visits `/tabs/profile`!	**5 / 6 Pass** (1 Warn, 0 Fail)

3. Master Defect & Remediation Priority Matrix (`QA-01` to `QA-14`)

USER & DEVELOPER ACTION PLAN

ID	SEVERITY	AFFECTED FILE(S) & LINES	ROOT CAUSE SUMMARY	EXACT FIX SPECIFICATION
QA-01	P0 CRITICAL	`auth.interceptor.ts:52–66` `connectivity.service.ts:43`	Single failed/rate-limited request on `/tabs/map` flips global `isHostOffline$` to `true` even when `/api/v1/health` is online.	Require 2 consecutive gateway errors or verify via `checkHostHealth()` before showing the top Offline Mode banner.
QA-02	P1 HIGH	`map.page.ts:535–620` `place.service.ts:113–169`	Online `/api/v1/places` query returns paginated subset (`limit=100`), replacing the 2,673 local OSM markers on pan/filter.	Merge live API results with the 2,673-item `osm_places.json` dataset in `PlaceService` so all 2,673 pins remain searchable and filterable.
QA-03	P0 CRITICAL	`suggested-wizard.page.html:31–56` `custom-builder.page.html:80–106`	Plain text input for Destination with no autofill dropdown; `.quick-dest-row` clips `Gapan City` as `"G..."` and omits 26 of 32 LGUs.	Add shared combobox indexing all 32 Nueva Ecija LGUs + 2,673 landmarks with diacritic-insensitive search (`munoz` → `Muñoz`) and wrap quick chips (`flex-wrap: wrap`).
QA-04	P0 CRITICAL	`itinerary.service.ts:865, 1038–1043, 1086`	Hardcoded `time: '10:00 AM - 12:00 PM'` on all OSM places makes sequential slot assignment dead code; overnight sort creates backwards timestamps.	Remove static `place.time` override and compute sequential, non-overlapping windows (`09:30 AM`, `12:00 PM`, `03:00 PM`, `06:00 PM check-in`) *after* chronological/2-Opt stop ordering.
QA-05	P0 CRITICAL	`itinerary.service.ts:914–1008` `itinerary_generator.py:185–263`	`categoryKey.includes('food')` misses `'restaurant'` / `'cafe'`; Day 1 ignores single-category preferences; overflow schedules 3 mountains/day and repeats *Pantabangan Dam* across days.	Use `PREFERENCE_CATEGORY_MAP` in TS, enforce strict preference matching with radial expansion (`15km -> 35km`), cap mountain peaks at 1/day + 1 daily lunch slot, and deduplicate by normalized venue name.
QA-06	P1 HIGH	`custom-builder.page.ts:551–564, 928–960`	Step 1 maps `culture` and `resort` into `'Tourist Spots'`, which filters Step 2 to only `tourist_attraction` / `nature_park`, hiding museums, churches, and resorts.	Map `'culture' -> 'Culture & Heritage'`, `'resort' -> 'Resorts & Stays'`, or add a `'Recommended'` filter tab that unions all selected Step 1 preference categories.
QA-07	P1 HIGH	`custom-builder.page.ts:865–879`	`loadBrowsePlaces()` calls `getFeaturedPlaces(50)`, showing only 17 curated places in Step 2 instead of the 2,673-POI catalog.	Load places for the selected Step 1 municipality from `searchPlaces({ municipality_name, limit: 100 })` merged with `loadOsmPlaces()`.

ID	SEVERITY	AFFECTED FILE(S) & LINES	ROOT CAUSE SUMMARY	EXACT FIX SPECIFICATION
QA-08..14	P2 POLISH	`app.component.ts`, `profile.page.ts`, `itinerary-detail.page.ts:400`, `home.page.html`	`ne_default_budget` and `ne_theme_mode` not loaded on app/wizard init; `ne_custom_api_url` not read by services; UTC `new Date()` in `ItineraryDetailPage.addDay()`.	Load `ne_theme_mode` in `AppComponent.ngOnInit()`, read `ne_default_budget` in both wizards' form builders, use `connectivityService.activeApiBaseUrl`, and use `addDaysLocal()` in `ItineraryDetailPage`.

4. Developer Remediation Blueprints for Top QA Failures

DROP-IN CODE FIXES

4.1 Fix `QA-04` & `QA-05`: Canonical Preference Matching, Deduplication & Non-Overlapping Time Slots (`itinerary.service.ts`)

```
// Project/src/app/core/services/itinerary.service.ts — Replace lines 914–1096 in generateOfflineSuggestedItinerary()
const PREF_TO_CATEGORIES: Record<string, string[]> = {
  nature: ['nature_park', 'tourist_attraction'],
  food: ['restaurant', 'cafe'],
  culture: ['museum_cultural', 'church_religious'],
  shopping: ['shopping_market', 'convenience_store'],
  resort: ['resort_swimming', 'hotel_accommodation'],
};

const preferredCatKeys = new Set<string>();
prefs.forEach((p) => (PREF_TO_CATEGORIES[p] || [p]).forEach((c) => preferredCatKeys.add(c)));

const matchesUserPrefs = (place: OfflineCuratedPlace): boolean => {
  if (preferredCatKeys.size === 0) return true;
  if (preferredCatKeys.has(place.categoryKey)) return true;
  return prefs.some((pr) => (place.tags || []).some((t) => t.toLowerCase().includes(pr)));
};

const normalizeVenueKey = (name: string): string =>
  (name || '').toLowerCase().replace(/\b(viewdeck|viewpoint|resto|restaurant|cafe|resort|spa|villas|and|&)\b/g, '').replace(/[^a-z0-9

// Ensure strict cross-day deduplication by both ID and normalized venue name (fixes "George Point" + "Pantabangan Dam" duplicates)
const usedVenueKeys = new Set<string>();

// After selecting day stops and sorting overnight stays to the end of the day, assign sequential non-overlapping time windows:
const BASE_SLOTS = [9 * 60 + 30, 12 * 60, 15 * 60, 18 * 60]; // 09:30 AM, 12:00 PM, 03:00 PM, 06:00 PM
let cursorMins = 9 * 60 + 30;
items.forEach((item, idx) => {
  item.sort_order = idx + 1;
  const dur = item.duration_minutes || 120;
  const travelBuffer = idx > 0 ? Math.max(15, Math.round((item.travel_duration_seconds || 900) / 60)) : 0;
  const startMins = this.isOvernightItem(item)
    ? Math.max(17 * 60, cursorMins + travelBuffer)
    : Math.max(BASE_SLOTS[idx] ?? cursorMins, cursorMins + travelBuffer);
  const endMins = startMins + dur;
  item.planned_time = `${this.formatMins12Hr(startMins)} - ${this.formatMins12Hr(endMins)}`;
  cursorMins = endMins;
});
```

4.2 Fix `QA-06`, `QA-07` & `QA-14C`: Custom Builder Step 2 Catalog & Global Settings Sync

```
// 1. Project/src/app/features/itinerary/custom-builder/custom-builder.page.ts (Fix QA-06 & QA-07)
public proceedToBuilder(): void {
  if (this.form.invalid) { this.form.markAllAsTouched(); return; }
  this.syncDaysWithDateRange();
  // Map single selected preference accurately; if multiple preferences selected, keep 'All' and sort matching categories first!
  if (this.selectedPreferences.size === 1) {
    const onlyPref = Array.from(this.selectedPreferences)[0];
    const singleMap: Record<string, string> = {
      food: 'Food', nature: 'Tourist Spots', culture: 'Culture & Heritage',
      resort: 'Resorts & Stays', shopping: 'Shopping',
    };
    this.selectedCategory = singleMap[onlyPref] || 'All';
  } else {
    this.selectedCategory = 'All';
  }
  this.loadBrowsePlaces();
  this.activeView = 'browse';
}

public loadBrowsePlaces(): void {
  this.isBrowseLoading = true;
  const dest = (this.form?.get('destination')?.value || '').trim();
  const muniParam = dest && !dest.toLowerCase().includes('nueva ecija') ? dest : undefined;
  this.placeService.searchPlaces({ municipality_name: muniParam, limit: 150 }).subscribe({
    next: (res) => {
      this.allCuratedPlaces = res.items?.length ? res.items : this.placeService.getLoadedPlaces(muniParam);
      this.applyBrowseFilter();
      this.isBrowseLoading = false;
    },
  });
}
```

```
// 2. Project/src/app/app.component.ts & Wizard Constructors (Fix QA-14C: Apply Saved Theme & Default Budget on Boot)
// In AppComponent.ngOnInit():
if (typeof localStorage !== 'undefined' && localStorage.getItem('ne_theme_mode') === 'dark') {
  document.body.classList.add('dark');
}
// In SuggestedWizardPage & CustomBuilderPage constructor:
const savedDefaultBudget = Number(typeof localStorage !== 'undefined' ? localStorage.getItem('ne_default_budget') : 0) || 3000;
```